PSG COLLEGE OF TECHNOLOGY

ERICSSON RESEARCH INDIA

COLLABORATIVE ROBOT HACKATHON

REPORT ON

AI POWERED VISION TASK FOR PLACEMENT OF GOODS

TEAM : INGENIX

Jayasri Rani S – 23Z328

Samyuktha Sri A – 23L258

Sudharsan M – 23R246

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SYNOPSIS

The rapid expansion of warehouse automation has created a growing demand for intelligent robotic systems capable of perceiving their environment and collaborating to perform complex manipulation tasks. Traditional warehouse robots often operate independently and rely on predefined rule-based programming, which limits adaptability and scalability. This project proposes an AI-powered vision-based collaborative robotic system for intelligent placement of goods in a warehouse environment. The system employs two fixed-base 3-degree-of-freedom robotic manipulators equipped with electromagnetic grippers to collaboratively pick, transfer, and place objects. A single overhead camera observes a predefined workspace of 40 × 40 cm and captures visual data for object detection. A custom-trained YOLOv8 model is used to detect and classify objects as either cube or cuboid, using a dataset labeled with Label Studio. Based on the detected object type, the system dynamically selects an appropriate manipulation strategy. For cube-shaped objects, a single robot performs the pickup operation and hands over the object to the second robot, which completes the placement task. For cuboid objects, both robots collaboratively lift and place the object. Pixel coordinates obtained from the vision system are calibrated to real-world coordinates and converted into joint angles using inverse kinematics. Control commands are transmitted wirelessly to the robots using ESP32 microcontrollers and the ESP-NOW communication protocol.

CHAPTER 1

INTRODUCTION

Automation and robotics are increasingly used in modern warehouses to improve efficiency, accuracy, and productivity. Traditional robotic systems usually operate independently and follow predefined instructions, which limits their ability to handle complex tasks that require cooperation between multiple robots. Many real-world tasks, such as lifting large objects or transferring items between locations, require coordinated actions similar to human teamwork. Advancements in computer vision and artificial intelligence have enabled robots to perceive their environment and make intelligent decisions. Vision-based object detection models can identify objects and determine their positions, allowing robots to perform pick-and-place operations more effectively. When multiple robots work together, proper coordination, communication, and motion planning are essential to avoid collisions and ensure successful task completion.This project focuses on designing a vision-based collaborative robotic system using two robotic manipulators. An overhead camera detects objects in the workspace using a YOLO-based object detection model. Based on the detected object type, the system selects an appropriate manipulation strategy and computes the robot joint movements using inverse kinematics. Communication between the robots is achieved through ESP32 modules using ESP-NOW, enabling coordinated object handling.The proposed system demonstrates how multiple robots can collaborate efficiently to perform object manipulation tasks while maintaining minimal communication and scalable system architecture

1.1 PROBLEM STATEMENT

Design a collaborative robot system using atleast two robots to place a specific object at a target location.The task is decomposed into sub-tasks such as object localization, path planning, cooperative picking, and ordered placement.A Visual Language Action (VLA) model is used for perception-to-action reasoning, enables coordination and decision making.Robots communicate minimally to share routes, coordinate carrying, avoid collisions, and ensure correct placement order.

1.2 NEED FOR THE PROJECT

Modern warehouses are increasingly transitioning towards automation to meet the demands of high throughput, accuracy, and reduced operational costs. Conventional robotic systems used in warehouses are often limited to single-robot operations and predefined task sequences, which restrict their ability to handle objects of varying sizes and weights. Additionally, these systems lack adaptability when multiple robots are required to collaborate for handling complex objects. Human workers frequently perform cooperative tasks such as lifting large objects or transferring items between stations. Replicating such collaboration in robotic systems requires effective perception, coordination, and communication mechanisms. Vision-based systems combined with intelligent decision-making frameworks offer a promising solution to this challenge. The need for this project arises from the requirement to design a collaborative robotic system capable of autonomously identifying objects, deciding manipulation strategies, and coordinating multiple robots using minimal communication and sensing hardware. By

leveraging computer vision and multi-robot coordination, the system aims to enhance efficiency, flexibility, and scalability in warehouse environments.

1.3 OBJECTIVES

The primary objectives of this project are:

- To design and develop a vision-based system for detecting and classifying warehouse objects.
- To fabricate and control two fixed-base 3-DOF robotic manipulators capable of collaborative operation.
- To implement scenario-based decision logic for single-robot and multi-robot manipulation.
- To perform camera calibration and inverse kinematics for accurate robot control.
- To establish wireless communication between robots using ESP32 and ESP-NOW. 7
- To design a system architecture that supports future implementation of Multi-Agent Reinforcement Learning (MARL) and Visual-Language-Action (VLA) models.

1.4 OVERVIEW OF THE PROJECT

The proposed system consists of an overhead vision module, two robotic manipulators, and a wireless control framework. A single camera mounted above the workspace captures images of objects placed within a 40 × 40 cm area. The images are processed using a custom-trained YOLOv8 object detection model to identify the object type and its location. Once the object is detected, its pixel coordinates are transformed into real-world coordinates through camera calibration. These coordinates are then converted into joint angles using inverse kinematics equations specific to the 3-DOF robotic arms. The computed angles are transmitted wirelessly to the robots via ESP32 microcontrollers.

Depending on the object classification, the system selects one of two operational scenarios:

- Scenario 1: Single-robot pickup followed by object handover and placement by the second robot.
- Scenario 2: Cooperative pickup and placement by both robots. This approach ensures efficient handling of objects with different physical characteristics while minimizing system complexity.

1.5 PROPOSED METHODOLOGY

The proposed methodology follows a structured pipeline:

1. Visual Perception: The overhead camera captures images of the workspace. YOLOv8 processes the images to detect and classify objects.
2. Decision Making: Based on the detected object type (cube or cuboid), the system selects the appropriate manipulation strategy.
3. Coordinate Transformation: Pixel coordinates are calibrated to camera coordinates and mapped to robot coordinates.
4. Inverse Kinematics: Joint angles for each robot are computed to reach the target position.
5. Wireless Communication: The calculated joint angles are transmitted using ESP-NOW to the respective robots.
6. Collaborative Manipulation: Robots execute pick-and-place operations either individually or collaboratively.This methodology ensures accurate, efficient, and scalable collaborative manipulation suitable for warehouse automation.

CHAPTER 2

LITERATURE SURVEY

Warehouse automation has witnessed significant advancements with the integration of robotics, computer vision, and artificial intelligence. This chapter reviews existing research and technologies related to object detection, robotic manipulation, collaborative multi-robot systems, and learning-based coordination, which form the foundation of the proposed work.

2.1 VISION-BASED OBJECT DETECTION IN ROBOTICS

Object detection plays a crucial role in robotic manipulation tasks, enabling robots to perceive their environment and identify objects of interest. Traditional object detection techniques relied on handcrafted features such as Haar cascades, Histogram of Oriented Gradients (HOG), and template matching. While effective in constrained environments, these methods lacked robustness under varying lighting conditions, orientations, and occlusions. With the advent of deep learning, Convolutional Neural Networks (CNNs) have significantly improved object detection accuracy. Among these, the You Only Look Once (YOLO) family of algorithms has gained prominence due to its real-time detection capability. YOLO processes the entire image in a single forward pass, making it suitable for time-critical robotic applications. Recent versions such as YOLOv8 further enhance detection performance by improving feature extraction, anchor-free detection, and computational efficiency. These characteristics make YOLOv8 well-suited for embedded robotic systems where both accuracy and speed are essential.

2.2 OBJECT DETECTION FOR ROBOTIC PICK-AND-PLACE TASKS

Several studies have demonstrated the application of vision-based object detection in robotic pick-and-place operations. Vision-guided manipulation allows robots to estimate object position, orientation, and size before performing grasping actions. However, many existing systems focus on single-robot manipulation and assume uniform object properties. In real-world warehouse environments, objects vary significantly in shape and size, requiring adaptive handling strategies. Research has shown that combining object classification with task-specific manipulation strategies improves efficiency and reduces failure rates. Vision- driven decision-making enables robots to select appropriate grasping and placement behaviors based on detected object characteristics. The proposed system builds upon these concepts by classifying objects as cubes or cuboids and dynamically selecting single-robot or multi-robot manipulation strategies.

2.3 ROBOTIC MANIPULATORS AND INVERSE KINEMATICS

Robotic manipulators are widely used in industrial automation due to their precision and repeatability. Fixed-base robotic arms with limited degrees of freedom are commonly employed in structured environments such as assembly lines and warehouses. 10 Inverse kinematics (IK) is a fundamental problem in robotics, involving the computation of joint angles required to position the end-effector at a desired location. Analytical IK solutions are computationally efficient and suitable for real-time control in low-DOF manipulators. Many studies emphasize the importance of accurate kinematic modeling and calibration to ensure precise motion execution. In vision-based robotic systems, the transformation of image coordinates to robot joint angles requires careful calibration and coordinate

mapping. Errors in calibration can significantly affect grasping accuracy. Therefore, camera calibration and IK form essential components of reliable robotic manipulation systems.

2.4 MULTI-ROBOT SYSTEMS AND COLLABORATION

Multi-robot systems offer improved scalability, redundancy, and efficiency compared to single- robot setups. Collaborative robots can share tasks, distribute workload, and handle objects that exceed the capabilities of individual robots. Research in this area has explored task allocation, coordination, and communication strategies among multiple robots. Traditional multi-robot systems often rely on centralized controllers and predefined task assignments. While effective in controlled environments, these approaches lack adaptability when task conditions change dynamically. Minimal communication strategies have been proposed to reduce system complexity and improve robustness. The proposed work adopts a lightweight coordination approach, where robots collaborate based on shared task context and minimal communication using wireless protocols. This approach ensures efficient coordination without excessive computational overhead.

2.5 WIRELESS COMMUNICATION IN ROBOTIC SYSTEMS

Reliable communication is essential for coordinated multi-robot operation. Common communication methods include Wi-Fi, Bluetooth, and wired connections. However, these methods may introduce latency, complexity, or scalability limitations. The ESP-NOW protocol, developed for ESP32 microcontrollers, enables low-latency, peer- to-peer wireless communication without requiring a centralized network. Studies have shown that ESP-NOW is suitable for real-time robotic coordination due to its fast transmission speed and low power consumption. By utilizing ESP-NOW, the proposed system achieves efficient communication between the main controller and individual robots, enabling synchronized execution of collaborative tasks.

2.6 VISUAL-LANGUAGE-ACTION (VLA) MODELS

Visual-Language-Action (VLA) models represent an emerging research direction that integrates visual perception, language understanding, and action execution. These models enable robots to interpret high-level task descriptions and translate them into executable actions based on visual input. VLA models offer the potential to replace rigid rule-based systems with flexible, human- interpretable task planning mechanisms. In warehouse automation, VLA models can enable dynamic task decomposition, adaptive decision-making, and improved human-robot interaction. While VLA models are still under active research, their integration with vision-based multi- robot systems presents a promising direction for future intelligent automation solutions.

2.7 SUMMARY OF LITERATURE SURVEY

From the literature review, it is evident that:

- Vision-based object detection using deep learning is essential for robotic manipulation.
- YOLO-based models provide real-time detection suitable for robotic applications.
- Collaborative multi-robot systems improve efficiency and handling capability.
- Wireless communication protocols such as ESP-NOW enable low-latency coordination.

However, there exists a gap in practical, low-cost, vision-based collaborative robotic systems that dynamically adapt manipulation strategies based on object characteristics.

The proposed work aims to address this gap by integrating vision-based detection, scenario-based coordination.

CHAPTER 3

PROPOSED WORK

This chapter presents the proposed collaborative robotic system designed for intelligent pick-and-place operations in a warehouse-like environment. The system integrates computer vision, robotic manipulation, wireless communication, and coordinated multi-robot control to perform autonomous object handling tasks.

An overhead camera is used to monitor the workspace and detect objects using a custom-trained YOLOv8 model. Based on the detected object type, the system selects an appropriate manipulation strategy and coordinates two robotic manipulators to perform the task. The system uses inverse kinematics for motion computation and ESP32 microcontrollers with ESP-NOW communication for wireless coordination between robots.

The following sections describe the system architecture, workflow, hardware components, software framework, and technical specifications used for implementing the proposed collaborative robotic system.

3.1 OVERVIEW

The proposed system is an AI-powered vision-based collaborative robotic framework designed for intelligent pick-and-place operations in a warehouse-like environment.

- The system integrates computer vision, robotic manipulation, wireless communication, and coordinated task execution to achieve autonomous object detection and placement.
- The overall architecture consists of two fixed-base 3-DOF robotic manipulators, an overhead vision module, and a wireless control framework.
- A single top-view camera continuously monitors a 40 × 40 cm workspace, minimizing occlusion and simplifying coordinate mapping between camera and robot frames.
- Captured images are processed using a custom-trained YOLOv8 object detection model developed with the Ultralytics framework.
- The model is trained to detect two object categories: Cube and Cuboid, using 100 manually labeled images annotated with Label Studio.
- The trained model provides real-time bounding box detection and object classification.
- Based on the detected object class, the system dynamically selects one of two manipulation strategies: Scenario 1 – Single-Robot Handover-Based Placement (for Cube detection) Scenario 2 – Dual-Robot Cooperative Placement (for Cuboid detection)
- In Scenario 1, Robot 1 picks up the object using an electromagnetic gripper, transfers it to a predefined handover position, and Robot 2 places it at the designated drop location.
- In Scenario 2, both robots simultaneously approach the object, collaboratively lift it using synchronized motion, transport it together, and place it in a coordinated manner.
- Each robotic arm is fabricated using 3D-printed components and operates with servo actuated joints for precise angular control.
- The 3-DOF configuration simplifies inverse kinematic computation while maintaining sufficient flexibility for planar manipulation.
- An electromagnetic gripper mounted at the end-effector enables secure grasping of metallic objects and is digitally controlled by an ESP32 microcontroller.

- The control system consists of three ESP32 microcontrollers: one Main ESP32 (central coordinator) and two Robot ESP32 units.
- Pixel coordinates from the YOLO model are converted into real-world coordinates through camera calibration and then transformed into respective robot coordinate frames.
- Analytical inverse kinematics is applied to compute required joint angles within servo 13 mechanical limits.
- The Main ESP32 transmits motion commands wirelessly using the ESP-NOW protocol, ensuring low-latency, peer-to-peer communication without a Wi-Fi router.
- The integrated system enables synchronized multi-robot coordination, intelligent scenario selection, and efficient collaborative object placement within a compact workspace

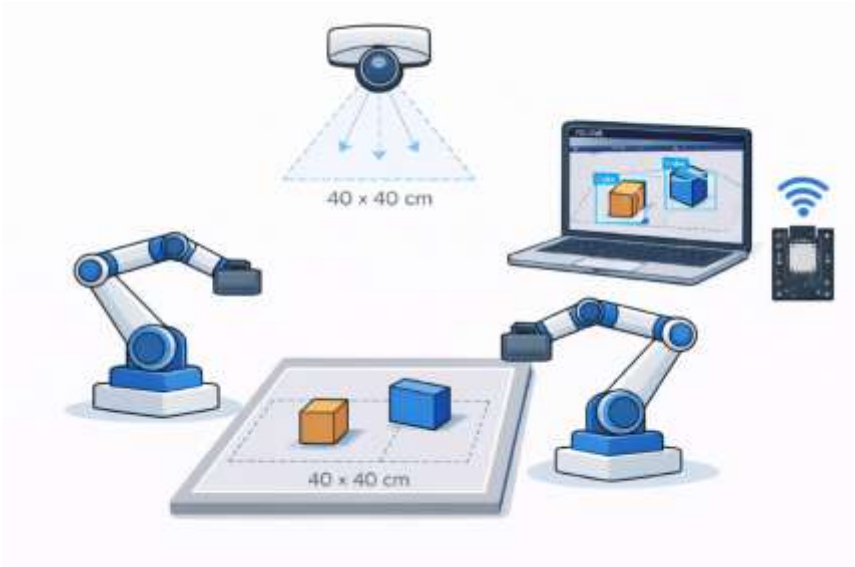


Fig 3.1 : System Implementation

3.2 SYSTEM FLOW DIAGRAM

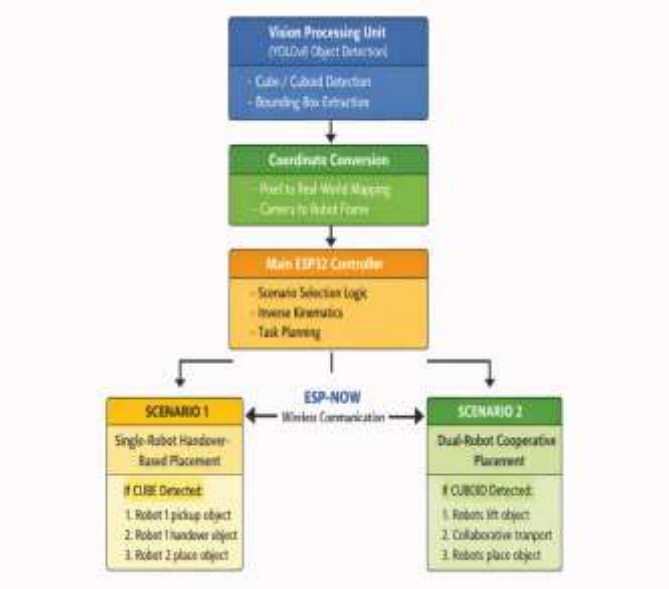


Fig 3.1:System Flow Diagram

3.3 HARDWARE USED

Table 3.1 List Of Hardware Used

S.NO	COMPONENT	FUNCTION
1	Web Camera	Captures real-time images of the 40 × 40 cm workspace for object detection.
2	24V SMPS Power Supply	Provides stable and regulated power to the entire robotic system.
3	Buck Converter	Steps down 24V to required lower voltage levels for ESP32 and other components.
4	ESP32 (3 Units)	One acts as Main Controller (central coordinator); two control individual manipulators.
5	Relay Module	Controls switching of high-power devices such as the electromagnetic gripper.
6	Electromagnetic (EM) Gripper	Picks and holds metallic objects using electromagnetic attraction.
7	Two 3-DOF Robotic Manipulators	Perform pick-and-place operations and collaborative manipulation.
8	Servo Motors (6 Units – 3 per manipulator)	Provide precise angular movement and joint control for robotic arms.

3.4 SOFTWARE USED

Table 3.2 List Of Software Used

S.NO	COMPONENT	FUNCTION
1	YOLOv8 (Ultralytics Framework)	Performs real-time object detection and classification (Cube / Cuboid) from captured images.
2	Label Studio	Used for annotating and labeling the training dataset images for YOLOv8 model training.
3	OpenCV (Python)	Used for image processing, frame acquisition, coordinate extraction, and camera interfacing.
4	Inverse Kinematics (IK) Algorithm	Computes the required joint angles for the 3-DOF robotic manipulators based on target position.
5	Python Programming	Implements the vision processing pipeline and integrates YOLOv8 with coordinate mapping.
6	Embedded C/C++ for ESP32 (PlatformIO in VS Code)	Used for motion control, servo operation, gripper control, and ESP-NOW wireless communication between controllers.

3.5 SPECIFICATIONS

3.5.1 VISION MODULE

Table 3.3 Specifications of Vision Module

S.NO	PARAMETER	SPECIFICATION
1	Camera Model	Zebronics Zeb-Crystal Pro Web Camera
2	Image Sensor	CMOS Sensor
3	Lens Type	3P Quality Lens
4	Resolution	640 × 480 Pixels
5	Frame Rate	30 FPS
6	Interface	USB 2.0
7	Operating Voltage	5V DC
8	Power Consumption	~2–3 W
9	Maximum Focal Length	2 meters

10	Aperture	f/2
11	Workspace Coverage	40 cm × 40 cm Area
12	Detection Model	YOLOv8 (Ultralytics Framework)
13	Dataset Annotation Tool	Label Studio
14	Image Processing Library	OpenCV (Python)
15	Detection Output	Object Class + Bounding Box + Pixel Coordinates

3.5.2 PICK AND PLACE MODULE

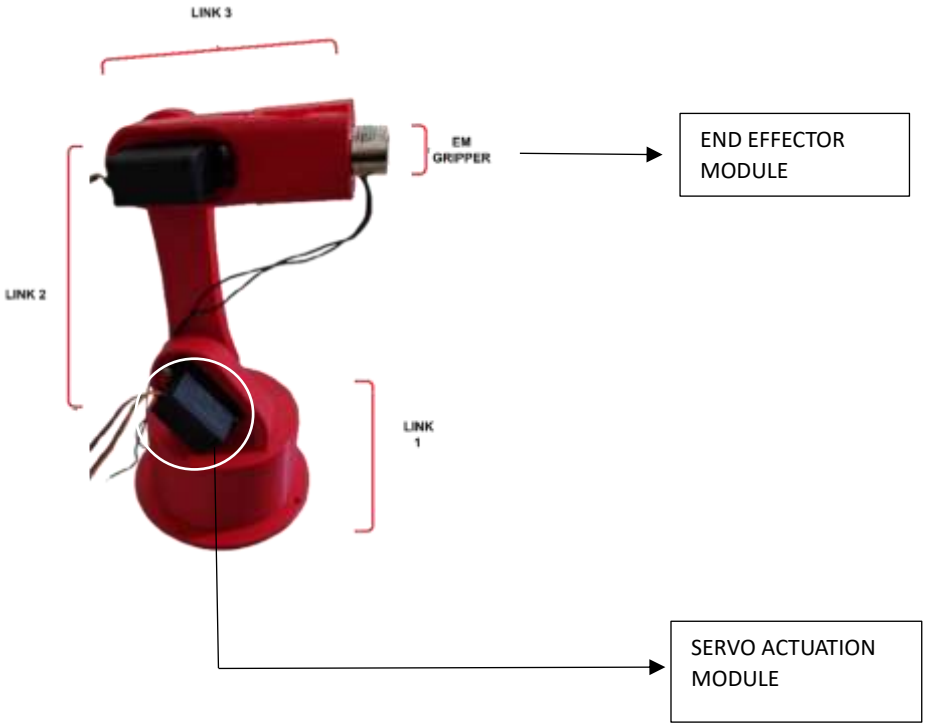


Fig 3.2 : 3-DOF Robotic Manipulator with Highlighted Structural Dimensions

Table 3.4 List of Robot Specifications

S.NO	PARAMETER	SPECIFICATION
1	Robot Type	Fixed-base Collaborative Robotic Manipulator
2	Number of Manipulators	2
3	Degrees of Freedom	3 DOF per manipulator
4	Joint Type	Revolute Joints
5	Joint Structure	Base Joint, Shoulder Joint, Arm Joint
6	Kinematic Model	Planar Articulated Manipulator
7	Motion Control	Joint Angle Control
8	Motion Computation	Analytical Inverse Kinematics
9	Workspace Type	Planar Workspace
10	Workspace Dimension	40 cm × 40 cm
11	Fabrication Method	3D Printed Components

12	Structure Type	Rigid Mechanical Structure
13	Rotation Constraint	0° – 180° Mechanical Limits
14	Payload Capacity	Up to 3 kg
15	Manipulation Modes	Single-Robot Handover & Dual-Robot Collaboration
16	Positioning Accuracy	±1–2 mm (Practical Tolerance)
17	Application	Autonomous Pick-and-Place Operation

Table 3.5 Specifications of End Effector Module

S.NO	PARAMETER	SPECIFICATION
1	Gripper Type	Electromagnetic (EM) Gripper
2	Rated Voltage	24V DC
3	Rated Current	0.5–1 A
4	Power Consumption	~24 W
5	Control Method	Relay Module
6	Control Signal	Digital GPIO from ESP32
7	Object Type	Metallic Objects

Table 3.6 Specifications of servo Actuation Module

S.NO	PARAMETER	SPECIFICATION
1	Actuator Type	High Torque Servo Motor
2	Total Servos	6 (3 per robotic manipulator)
3	Degrees of Freedom Controlled	3 DOF per robot
4	Joint Positions	Base Joint, Shoulder Joint, Arm Joint
5	Control Method	PWM Signal from ESP32
6	Operating Voltage	6V DC
7	Rotation Range	0° – 180°
8	Maximum Current	2.5 A (Stall Current)
9	Control Interface	ESP32 GPIO Pins
10	Actuation Method	Direct Servo Drive
11	Motion Resolution	Angular Position Control
12	Application	Robotic Arm Joint Movement

Table 3.7 Specifications Of Power Supply Module

S.NO	PARAMETER	SPECIFICATION
1	Input Supply	230V AC, 50 Hz
2	Main Power Supply	24V DC SMPS
3	SMPS Rated Current	5 A
4	Maximum SMPS Power	120 W
5	Power Distribution Method	DC Power Distribution Bus
6	Buck Converter 1	24V → 6V DC Step-down Converter
7	Buck Converter 1 Current Rating	5 A
8	Buck Converter 2	24V → 5V DC Step-down Converter

9	Buck Converter 2 Current Rating	3 A
10	Servo Supply Voltage	6V DC (via Buck Converter)
11	Logic Supply Voltage	5V DC (via Buck Converter)
12	ESP32 Operating Voltage	3.3V Logic (Internally regulated from 5V)
13	EM Gripper Supply	24V Direct from SMPS
14	Relay Module Supply	5V DC
15	Total Servo Maximum Current	~15 A (6 Servos Combined Peak)
16	Grounding Scheme	Common Ground Configuration
17	Power Conversion Method	High-efficiency DC Buck Conversion
18	Estimated System Load	60–120 W (Dynamic Load)

Table 3.8 Specifications Of Communication Module

S.NO	PARAMETER	SPECIFICATION
1	Microcontroller	ESP32
2	Total Controllers	3 (1 Main + 2 Robot Controllers)
3	Communication Protocol	ESP-NOW
4	Frequency Band	2.4 GHz
5	Communication Type	Peer-to-Peer
6	ESP32 Supply Voltage	5V
7	Logic Voltage	3.3V
8	Average Current	150–250 mA
9	Peak Current	~500 mA
10	Communication Latency	Few milliseconds

CHAPTER 4

SYSTEM IMPLEMENTATION

This chapter describes the practical implementation of the proposed AI-powered vision-based collaborative robotic system. It explains the hardware fabrication, software development, vision model training, calibration process, and communication mechanism used to realize the system.

4.1 HARDWARE IMPLEMENTATION

The hardware implementation focuses on constructing reliable robotic manipulators and integrating them with control and communication units.

4.1.1 FABRICATION OF ROBOTIC MANIPULATORS

Two fixed-base 3-degree-of-freedom robotic manipulators were fabricated using 3D- printed components. The links and joints were designed to ensure sufficient reach within the 40 × 40 cm workspace. Each joint is actuated using servo motors to provide accurate angular positioning. 23 The base of each robot is rigidly fixed to the work surface to eliminate vibrations and ensure consistent motion during pick-and-place operations.



Fig 4.1 : 3D Printed Base, Link Arms, and Joint Components of the Proposed Robotic Manipulator



Fig 4.2 : Fully Assembled 3-DOF Robotic Manipulator with Mounted Servo Joints

4.1.2 INTEGRATION OF ELECTROMAGNETIC GRIPPER

An electromagnetic gripper is mounted at the end-effector of each robotic arm. The gripper is controlled using digital output signals from the ESP32 microcontroller. When energized, the gripper generates a magnetic field that securely holds metallic objects.The gripper provides fast actuation and simplifies mechanical design, making it suitable for repetitive pick-and-place tasks.

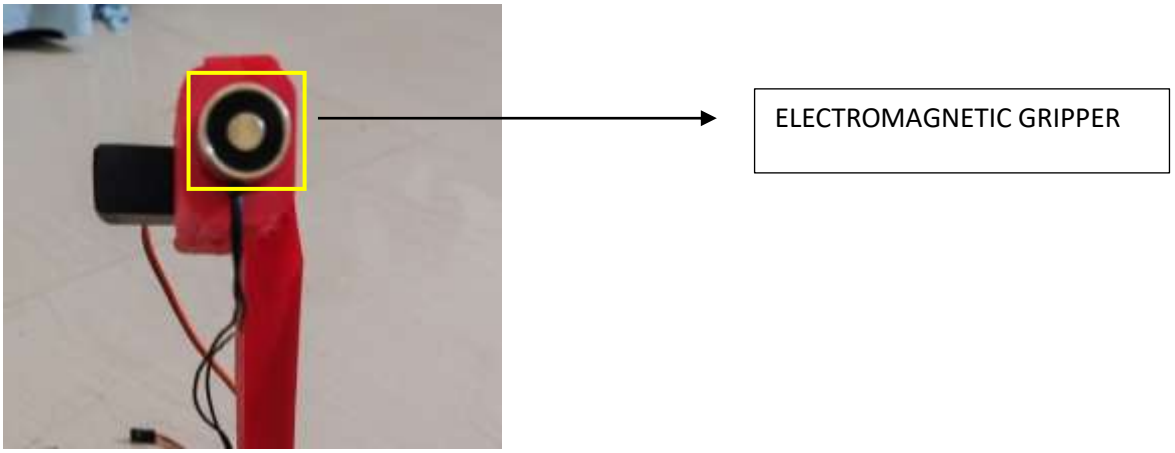


Fig 4.3 : Manipulator with EM gripper

4.2 CONTROL HARDWARE USING ESP 32

Each robotic arm is controlled by an individual ESP32 microcontroller, while an additional ESP32 functions as the main coordinator.

- Main ESP32: Handles decision-making, receives vision output, computes inverse kinematics, and sends control commands.
- Robot ESP32 Units: Receive joint angle commands and control servo motors and grippers.

The ESP32 platform was selected due to its processing capability, wireless features, and ease of integration with sensors and actuators.

4.3 SOFTWARE IMPLEMENTATION

The software implementation consists of vision processing, decision logic, kinematic computation, and communication modules.

4.3.1 VISION PROCESSING USING YOLOV8

The vision module is implemented in Python using the Ultralytics YOLOv8 framework. A custom dataset containing 100 images of cube and cuboid objects was prepared. Images were captured under varying orientations and lighting conditions within the workspace. Label Studio was used to annotate the dataset by drawing bounding boxes and assigning class labels. The dataset was then used to train the YOLOv8 model.

During execution, the trained model processes real-time camera frames and outputs:

- Object class (cube or cuboid)
- Bounding box coordinates
- Confidence score

1.F1–Confidence Curve:

The F1-score is the harmonic mean of Precision and Recall

$$F1 = (2 \times Precision \times Recall)/(Precision + Recall) \quad (4.1)$$

The F1–Confidence curve shows how detection quality varies with different confidence thresholds.

From the graph:

- Maximum F1-score ≈ **0.99 at confidence ≈ 0.53**
- Both **Cube and Cuboid curves overlap closely**
- Indicates balanced precision and recall

This indicates:

- At confidence threshold ≈ **0.53**, the model gives optimal performance.
- The model detects cube and cuboid objects with very high reliability.
- Minimal false positives and false negatives at this threshold.

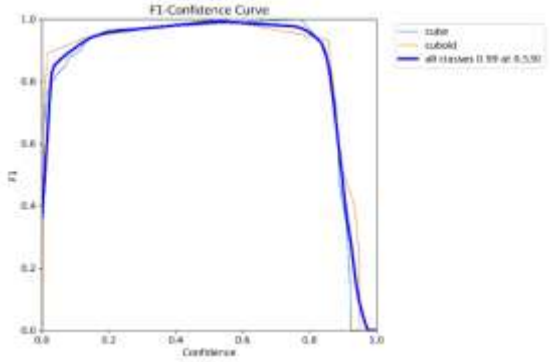


Fig 4.4 : F1 Score vs Confidence Curve for the Proposed Object Detection Model

2. Precision–Confidence Curve

Precision measures:

$$Precision = \frac{TruePositives}{TruePositives + FalsePositives} \quad (4.2)$$

It shows how accurate predictions are when the model declares an object

Interpretation from Graph

- Precision reaches **1.00 at confidence ≈ 0.799**
- Both classes approach near-perfect precision at high confidence.
- Cuboid slightly higher at low confidence.

This indicates:

- At higher confidence thresholds, almost **no false detections occur**.
- When the model predicts “Cube” or “Cuboid,” it is nearly always correct.
- Very suitable for robotic pick-and-place tasks where false detection is critical.

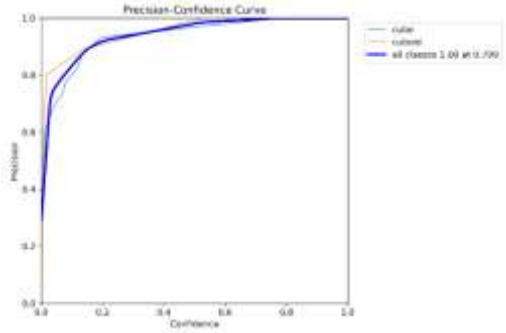


Fig 4.5 : Precision–Confidence Curve of the Trained YOLOv8 Model for Cube and Cuboid Detection

3. Precision–Recall Curve

his curve shows trade-off between precision and recall.

- Ideal curve → close to top-right corner.
- Area under curve = Average Precision (AP)

Model Result

- mAP@0.5 ≈ **0.995**
- Both cube and cuboid ≈ 0.995

This indicates

- The model almost perfectly separates cube and cuboid.
- Very high detection quality.
- Strong generalization within 40 × 40 workspace

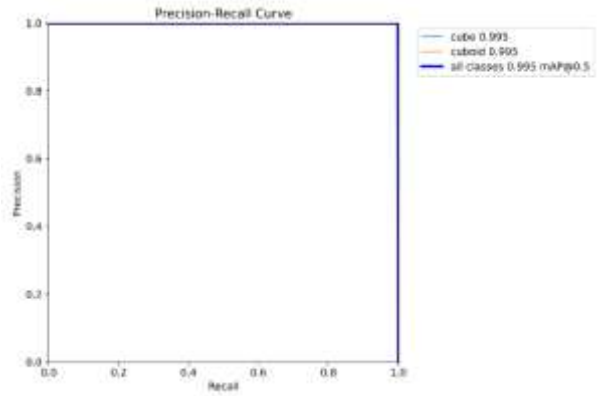


Fig 4.6 : Precision–Recall Curve Demonstrating Detection Performance of the Proposed Object Detection System

3. Recall–Confidence Curve

Recall measures:

$$Recall = \frac{TruePositives}{TruePositives+FalseNegatives} \quad (4.3)$$

It shows how many actual objects are correctly detected.

Interpretation

- Recall remains near **1.0 for lower confidence values**.
- Drops sharply near confidence 0.85–0.9.

This indicates

- At moderate confidence (≈0.5), the system detects nearly all cube and cuboid objects.
- If confidence threshold is too high, some objects may be missed.
- Therefore, 0.5–0.6 threshold is optimal for robotic operation.

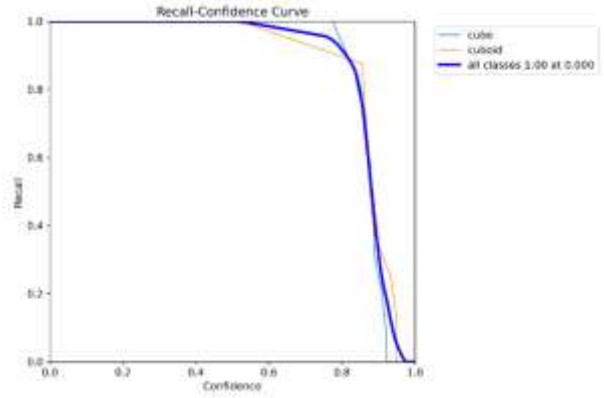


Fig 4.7 : Recall–Confidence Curve of the Proposed YOLOv8 Model

5. Training and Validation Performance Plots (10 Graphs)

These plots evaluate model learning behavior.

(1) train/box_loss

Represents bounding box regression loss.

- Decreasing steadily
- Indicates improving localization accuracy

Model learns object position accurately.

(2) train/cls_loss

Represents classification loss.

- Decreasing smoothly
- Indicates improved cube vs cuboid classification

Model differentiates shapes clearly.

(3) train/dfl_loss

Distribution Focal Loss for bounding box refinement.

- Decreases over epochs
- Shows improved bounding box precision

(4) metrics/precision(B)

Precision during training.

- Approaches near 1.0
- Stable

(5) metrics/recall(B)

Recall during training.

- Approaches near 1.0
- Stable performance
- Very few missed detections.

(6) val/box_loss

Validation bounding box loss.

- Slight fluctuations
- Overall decreasing trend

Good generalization.

(7) val/cls_loss

Validation classification loss.

- Decreases and stabilizes
- No major overfitting

Cube and cuboid separation consistent.

(8) val/dfl_loss

Validation localization refinement.

- Small oscillations
- Remains stable

Model robust to slight variation.

(9) metrics/mAP50(B)

Mean Average Precision at IoU = 0.5

- Approaches 0.99–1.00
- Excellent detection quality

(10) metrics/mAP50-95(B)

Mean Average Precision averaged across IoU 0.5–0.95

- Reaches ~0.7+
- Indicates strong bounding box tightness

Overall Model Performance Summary

For Cube and Cuboid Detection:

Table 4.1: Model Perfomance Summary

Metric	Value
F1 Score	~0.99
Precision	~1.00
Recall	~1.00
mAP@0.5	0.995
mAP@0.5–0.95	~0.70+

Conclusion for Vision System

The trained YOLOv8 model demonstrates:

- High precision in classifying cube and cuboid
- Excellent recall ensuring no object is missed
- Accurate bounding box localization
- Stable training without overfitting
- Suitability for real-time autonomous robotic pick-and-place

The performance curves confirm that the model is reliable for deployment in the 40 × 40 cm workspace for collaborative cobot handling.

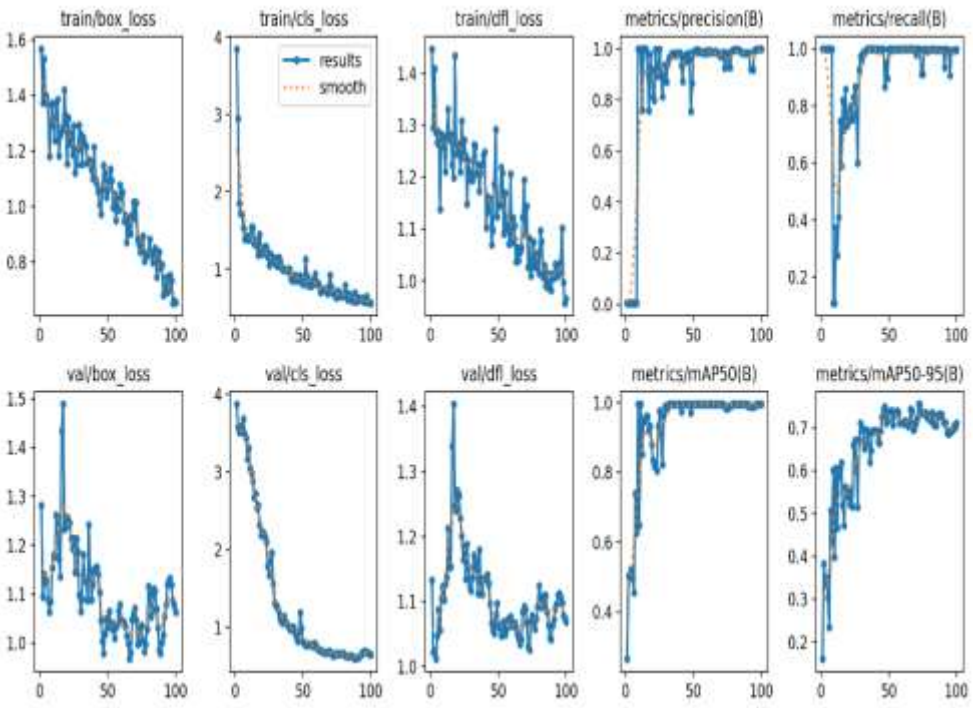


Fig 4.8 : Training and Validation Performance Curves of the Proposed YOLOv8 Model

4.3.2 SCENERIO SELECTION LOGIC

The object class detected by YOLOv8 determines the task execution strategy.



Fig 4.9 : Cube and cuboid object together

- If the object is classified as a **cube**, the system selects **Scenario 1**, where a single robot performs pickup and hands over the object.

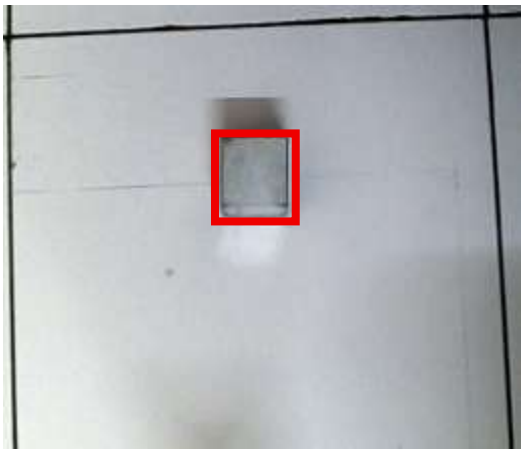


Fig 4.10 : Cube object in the respective workspace

- If the object is classified as a **cuboid**, the system selects **Scenario 2**, where both robots collaborate to lift and place the object.



Fig 4.11 : Cuboid object in the respective workspace

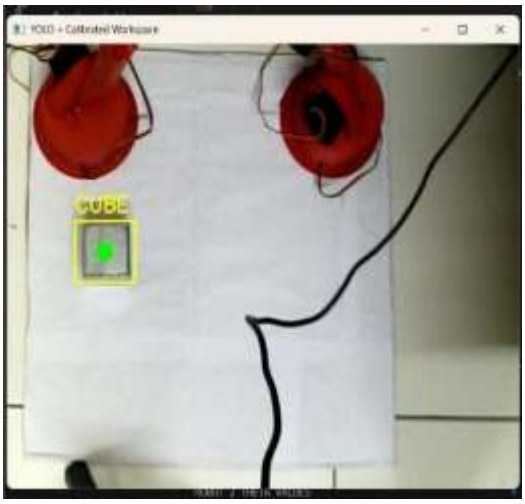


Fig 4.12 : Object detection using YOLO

4.4 CAMERA CALIBRATION AND WORKSPACE REPRESENATION

The YOLO model outputs object locations in pixel coordinates. These coordinates must be converted into real-world positions before robotic manipulation.



Fig 4.13 : Centralised camera for object detection

Physical Workspace Definition

The robotic system operates within a defined **open workspace of 40 cm × 40 cm**.
This workspace is positioned centrally between the two 3DOF articulated cobots.

- Workspace Type: Open planar workspace
- Physical Dimensions: 40 cm (X-axis) × 40 cm (Y-axis)
- Total Area: 1600 cm²
- Reference Frame: Cartesian coordinate system

The origin (0,0) is defined at one corner of the workspace, and coordinates increase positively along X and Y directions.

(0,40)



Fig 4.14 : Workspace calibration with dimension 40 x 40

Conversion of Workspace into Matrix Representation

To enable computational mapping and robotic manipulation, the physical workspace is discretized into a matrix.

Matrix Resolution Selection

For simplicity and direct mapping:

$$1 \text{ grid cell} = 1 \text{ cm} \times 1 \text{ cm}$$

Therefore:

$$40 \text{ cm} \rightarrow 40 \text{ cells}$$

The workspace is represented as:

$$40 \times 40 \text{ matrix}$$

Total grid cells:

$$40 \times 40 = 1600 \text{ discrete cells}$$

Purpose of Matrix Representation

The matrix representation is used for:

- Mapping pixel coordinates to real-world coordinates
- Identifying object position
- Determining robot reachability
- Grid-based occupancy analysis
- Collision avoidance and workspace constraint evaluation

This discretization simplifies coordinate transformation from image space to robotic space.

Pixel-to-Workspace Mapping

The overhead camera captures the workspace in pixel coordinates.

Let:

- Camera resolution = 1920 × 1080 pixels (example)
- Workspace region extracted from image

Scaling factors:

$$Scale_X = \frac{40\ \text{cm}}{Image\ Width\ in\ Pixels}$$

$$Scale_Y = \frac{40\ \text{cm}}{Image\ Height\ in\ Pixels} \quad (4.4)$$

Thus:

$$X_{cm} = X_{pixel} \times Scale_X \quad (4.5)$$

$$Y_{cm} = Y_{pixel} \times Scale_Y \quad (4.6)$$

This converts detected object center into Cartesian coordinates within the 40 × 40 matrix.

Object Dimensions Used in Project

The metallic test objects used in the project are:

Table 4.2 : Object type used with dimensions

Object Type	Dimensions
Cube	4 cm × 4 cm × 4 cm
Cuboid	8 cm × 4 cm × 4 cm

Grid Cell Occupancy by Objects

Since:

$$1 \text{ grid cell} = 1 \text{ cm}$$

Cube Occupancy

$$4 \text{ cm} \times 4 \text{ cm}$$

Cube occupies:

$$4 \times 4 = 16 \text{ grid cells}$$

Cuboid Occupancy

$$8 \text{ cm} \times 4 \text{ cm}$$

Cuboid occupies:

$$8 \times 4 = 32 \text{ grid cells}$$

The camera is calibrated with respect to the workspace dimensions (40 × 40 cm). A linear mapping technique is used to convert pixel coordinates into camera coordinates. The camera coordinate system is then aligned with the robot coordinate frames.

This calibration ensures accurate mapping between visual perception and physical motion.

4.5 INVERSE KINEMATICS AND MOTION CONTROL

Inverse kinematics calculations are implemented to convert target positions into joint angles for the robotic arms.

Given the desired end-effector position, analytical inverse kinematics equations are applied to compute joint angles for each of the three joints. Joint limits are enforced to prevent mechanical damage and ensure safe operation.



Fig 4.15 : Angles computed through inverse kinematics

4.5.1 SCENARIO 1: SINGLE-ROBOT PICKUP WITH HANDOVER

When a **cube** is detected:

- Robot 1 moves to the detected object location.
- The electromagnetic gripper is activated to pick up the object.
- Robot 1 transfers the object to a predefined handover position.
- Robot 2 receives the object and transports it to the destination location.
- The object is placed at the drop point.

This scenario minimizes unnecessary multi-robot involvement for lightweight objects.

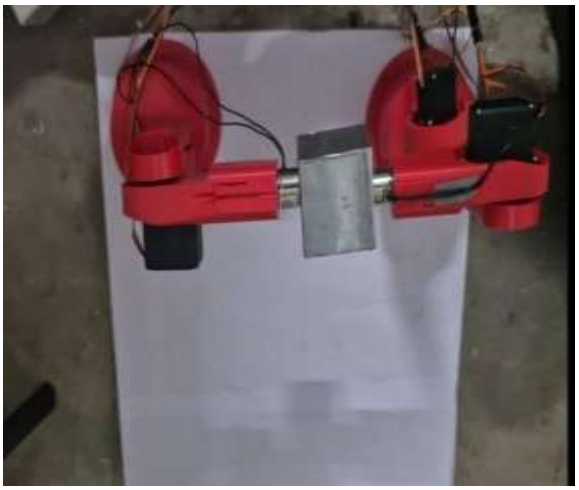


Fig 4.16 : Demonstration of scenerio 1

4.5.2 SCENARIO 2: DUAL-ROBOT COOPERATIVE PICKUP

When a **cuboid** is detected:

- Both robots move simultaneously to the object location.
- The object is lifted collaboratively using synchronized motion.
- The robots transport the object to the destination.
- The object is placed at the drop point in a coordinated manner.
- This approach ensures stable handling of larger objects requiring cooperative manipulation.

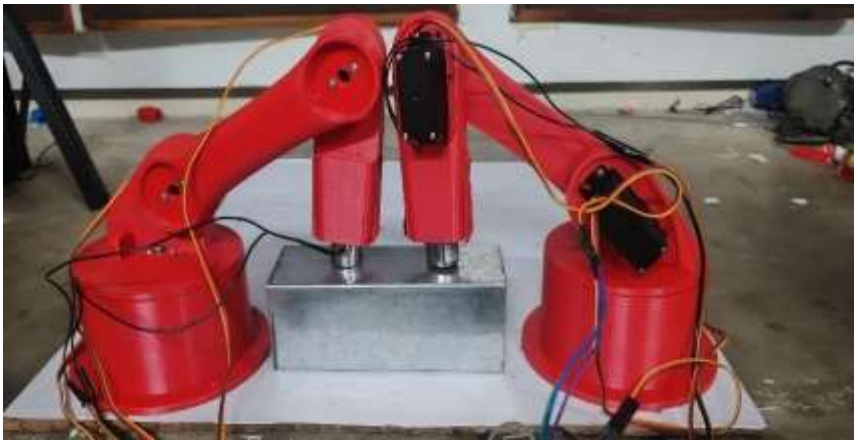


Fig 4.17 : Demonstration of scenerio 2

4.6 WIRELESS COMMUNICATION USING ESP-NOW

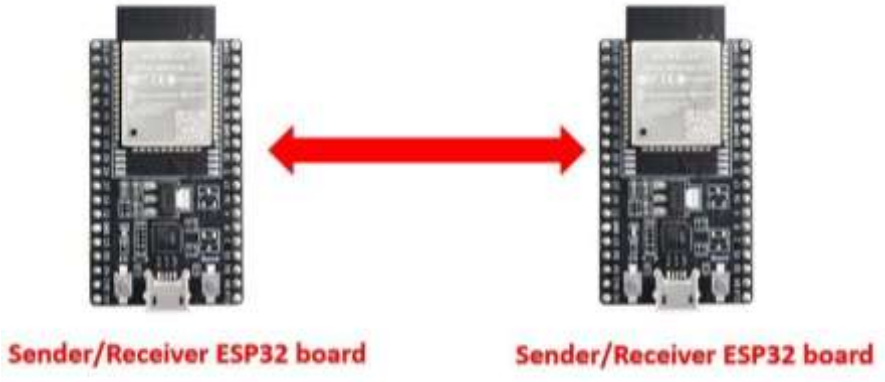


Fig 4.18 : Communication between ESP 32

Introduction to ESP-NOW

ESP-NOW is a **connectionless, peer-to-peer wireless communication protocol** developed by Espressif Systems for ESP32 and ESP8266 devices.

It operates:

- At **2.4 GHz Wi-Fi Physical Layer (IEEE 802.11 b/g/n PHY)**
- Without requiring a router
- Without TCP/IP stack
- With low latency and low power consumption

In the proposed collaborative robotic system, ESP-NOW is used to synchronize motion between Robot 1 and Robot 2 during pick-and-place and handover operations.

Network Architecture Used in the Project

The implemented architecture is:

Peer-to-Peer Architecture

- Robot 1 ESP32 ↔ Robot 2 ESP32
- Direct MAC address-based communication
- No access point required
- No TCP/IP stack required

Each ESP32 is configured in:

```
WiFi.mode(WIFI_STA);
```

ESP-NOW works at:

- MAC layer
- Bypasses IP layer
- Uses vendor-specific action frames

OSI Layer Working in ESP-NOW

ESP-NOW operates mainly in the lower layers of the OSI model:

Table 4.3 : OSI layer working in ESP NOW

OSI Layer	ESP-NOW Implementation
Application Layer	User-defined data (e.g.,"PICKUP_REACHED")
Transport Layer	Not used
Network Layer	Connectionless (No IP)
Data Link Layer (MAC)	802.11 Vendor-Specific Action Frame
Physical Layer	2.4 GHz Wi-Fi PHY

Physical Layer (PHY)

- Frequency Band: 2.4 GHz
- Standard: IEEE 802.11b/g/n
- Modulation: DSSS / OFDM
- Maximum Transmission Power: +20 dBm
- Data Rate: Up to 1 Mbps (ESP-NOW optimized)

MAC Layer Operation

ESP-NOW uses : Vendor-Specific Action Frames

These are special IEEE 802.11 management frames.

Frame Type: IEEE 802.11 Action Frame

Vendor OUI (Espressif):18:FE:34

MAC addressing is used for direct peer identification.

In code:

```
uint8_t ESP1_MAC[] = {0x78,0x1C,0x3C,0xA4,0xCA,0x8C};
```

Each message is sent directly to this MAC address.

ESP-NOW Data Frame Size

According to Espressif documentation:

- Maximum payload size = **250 bytes**
- No IP header
- Low overhead

Your system uses very small packets (< 30 bytes), ensuring:

- Minimal delay
- Fast synchronization
- Reliable communication

General ESP-NOW Frame Structure (As per Espressif)

| MAC Header | Category Code | OUI | Random Value | Data | FCS |

Table 4.4:ESP NOW frame structure

Field	Size
Frame Control	2 bytes
Duration	2 bytes
Destination MAC	6 bytes
Source MAC	6 bytes
BSSID	6 bytes
Sequence Control	2 bytes
Category Code	1 byte
OUI	3 bytes
Random Value	4 bytes
Vendor-Specific Content	0–250 bytes
Frame Check Sequence	4 bytes

Packet Structure Used in Code – Scenario 1

Code sends string-based messages:

Example:

```
esp_now_send(ESP1_MAC, (uint8_t*)"PICKUP_REACHED", 14);
```

Scenario 1 Packet Structure

Table 4.5: Application Layer Payload

Field	Type	Size
Command String	char[]	14–20 bytes

Examples:

Table 4.6: Message transferred between Robots

Message	Meaning
"PICKUP_REACHED"	Robot 1 reached pickup
"HANDOVER_REACHED"	Robot 1 reached handover
"OBJECT_RELEASED"	Object demagnetized
"MAGNETISING_ROBOT2"	Robot 2 magnet ON
"MOVING_TO_DROP"	Robot 2 moving to drop

Actual Data Payload Example:

PICKUP_REACHED

Byte representation: 50 49 43 4B 55 50 5F 52 45 41 43 48 45 44

Size: 14 bytes

Total ESP-NOW payload: 14 bytes (well below 250-byte limit)

Scenario 2 Packet Structure

Table 4.7 : Application Layer Payload

Field	Value	Size
Command String	IN_SYNC	7 bytes

Actual Data Payload Example:

IN_SYNC

Byte representation: 49 4E 5F 53 59 4E 43

Size:7 bytes

Total ESP-NOW payload:7 bytes (well below 250-byte limit)

SCENERIO 1 : SINGLE-ROBOT HANDOVER-BASED PLACEMENT

ROBOT 1 ROBOT 2

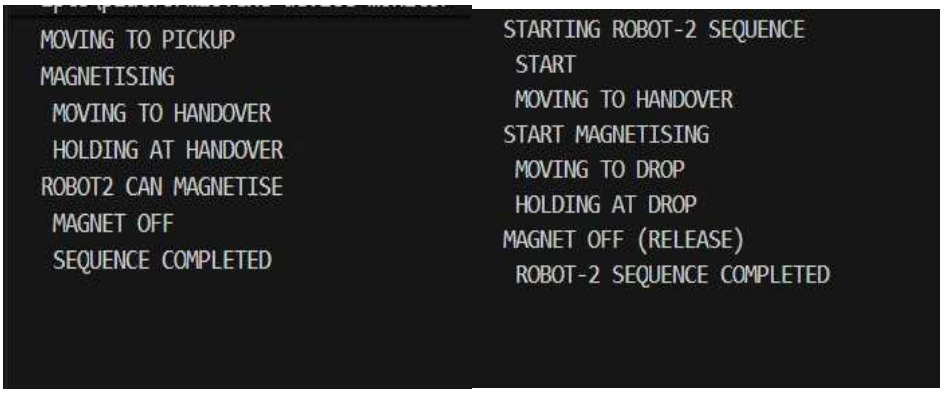


Fig 4.19 : Communication between robots during scenerio 1

SCENERIO 2 : DUAL ROBOT COOPERARIVE PLACEMENT

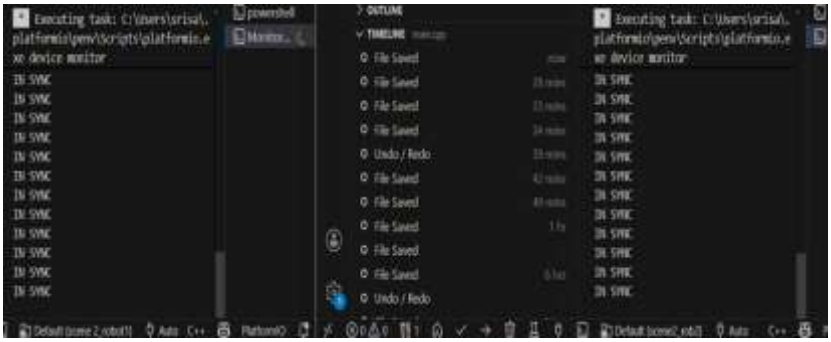


Fig 4.20 : Communication between robots during scenerio

4.7 IMPLEMENTATION CHALLENGES

During implementation, several challenges were encountered:

- Calibration errors due to camera alignment
- Synchronization between robots during cooperative lifting
- Servo motor limitations and mechanical tolerances
- Wireless communication delays under interference

These challenges were addressed through iterative testing and parameter tuning.

4.8 DIGITAL TWIN IMPLEMENTATION

The proposed AI-powered vision-based collaborative robotic system is physically implemented using two fixed-base 3-DOF manipulators equipped with electromagnetic grippers and coordinated using ESP32-based wireless communication. The Digital Twin simulation platform represents a real-time virtual replica of this physical system, mirroring its kinematic behavior, magnetization state, collaborative modes, and system performance metrics.

In the physical implementation, object detection is performed using a custom-trained YOLOv8 model. Based on classification (cube or cuboid), the system selects either:

- **Handover Mode (Single-Robot Transfer)**
- **Coordinated Mode (Dual-Arm Lift)**

In the Digital Twin interface, these same operational protocols are visually represented under the “Active Protocol” section. The handover mode simulates synchronized transfer between Robot A and Robot B, while the Coordinated Mode represents dual-arm lifting with simultaneous trajectory execution. This ensures that the simulation logic directly corresponds to the physical task execution logic implemented in the proposed system.



Fig 4.21: Implementation of digital twin

4.8.1 MAGNETIZATION AND GRIPPER STATE REPRESENTATION

In the physical system, the electromagnetic gripper is activated through digital signals from the ESP32 controller. Magnetization enables secure gripping of metallic objects during pickup and transfer operations.

The Digital Twin simulation reflects this behavior by displaying real-time magnet states such as:

- **MAGNET: ACTIVE**
- **MAGNET: IDLE**

These indicators visually represent when the gripper is energized or deactivated. In **Handover Mode**, the simulation clearly shows one robot's magnet active during pickup, followed by controlled deactivation during transfer, replicating the real-world sequence. This direct visualization improves validation of gripping logic and synchronization timing.

4.8.2 JOINT ANGLE MONITORING AND KINEMATIC FEEDBACK

In the proposed hardware setup, inverse kinematics computes joint angles (shoulder, elbow, and wrist), which are transmitted via **ESP-NOW** to each robot. These angles determine the physical movement of the servo motors.

The Digital Twin enhances this by displaying live telemetry for:

- **Shoulder angle**
- **Elbow angle**
- **Wrist angle**
- **Power percentage**

For example, Robot A and Robot B angle values are shown dynamically during simulation. This enables verification of inverse kinematics outputs before physical actuation. The simulation also confirms whether the kinematic solver is active, ensuring that joint computations are being applied in real time.

4.8.3 COLLISION AVOIDANCE AND SAFETY VALIDATION

In the real system, safety depends on calibration accuracy and synchronized movement. Any error in inverse kinematics or timing may result in misalignment or collision.

The Digital Twin incorporates a **Collision Avoidance indicator**, confirming that simulated trajectories are validated against workspace constraints. This acts as a preventive safety layer before deploying commands to physical robots.

4.8.4 PERFORMANCE METRICS AND ANALYTICAL EVALUATION

The proposed system experimentally measures manipulation accuracy and task completion. The Digital Twin extends this by providing analytical performance metrics including:

- **Payload tracking (X, Y, Z coordinates)**
- **System efficiency (e.g., 95%)**
- **Cycle time (~2.0 seconds)**
- **Precision (±0.04 mm)**
- **Uptime percentage**
- **Reach capability**

These metrics provide quantitative evaluation of collaborative task performance in a visual and analytical format, complementing the qualitative validation performed in physical experiments.

CHAPTER 5

EXPERIMENTAL SETUP AND RESULTS

This chapter describes the experimental setup used to validate the proposed AI-powered vision-based collaborative robotic system and discusses the observed results. The system was tested under controlled conditions to evaluate its ability to detect objects, select appropriate task execution strategies, and perform collaborative pick-and-place operations.

5.1 EXPERIMENTAL SETUP

The experimental setup was designed to closely resemble a simplified warehouse environment while maintaining control over variables affecting system performance.

5.1.1 WORKSPACE CONFIGURATION

A square workspace of dimensions 40 × 40 cm was defined and monitored using a single overhead camera. The camera was mounted at a fixed height to ensure complete coverage of the workspace and minimize perspective distortion. The two robotic manipulators were fixed on opposite sides of the workspace to enable both individual and collaborative manipulation of objects.

5.1.2 OBJECT USED FOR TESTING

Two types of metallic objects were used for experimentation:

- Cube-shaped objects
- Cuboid-shaped objects

These objects were selected to evaluate the system's ability to dynamically adapt manipulation strategies based on object geometry.

5.1.3 SYSTEM COMPONENTS

The experimental setup consisted of:

- Overhead camera for vision input
- YOLOv8-based object detection module
- Main ESP32 controller
- Two robot-specific ESP32 controllers
- Electromagnetic grippers mounted on each robot

The system was powered using regulated power supplies to ensure stable operation.

5.2 TEST SCENERIOS

The system was evaluated under two primary test scenarios corresponding to the implemented task execution strategies.

5.2.1 SCENERIO 1 SINGLE-ROBOT PICKUP WITH HANDOVER

In this scenario, a cube-shaped object was placed at a random location within the workspace. The vision system successfully detected the object and classified it as a cube.

Robot 1 navigated to the detected position, activated its electromagnetic gripper, and lifted the object. The object was then transferred to a predefined handover position. Robot 2 received the object and placed it at the designated drop location. This scenario validated the system's ability to perform sequential multi-robot collaboration using a handover mechanism.

5.2.2 SCENERIO 2 DUAL-ROBOT COOPERATIVE PICKUP

For cuboid-shaped objects, the system selected the cooperative manipulation strategy. Both robots moved simultaneously toward the object and engaged their electromagnetic grippers. The robots lifted the object together and transported it to the destination location. Coordinated motion ensured stable handling of the object throughout the process. This scenario demonstrated effective synchronization and collaboration between robots.

5.3 OBSERVED RESULTS

The experimental results were evaluated qualitatively based on system behavior and task completion.

5.3.1 OBJECT DETECTION PERFORMANCE

The YOLOv8 model successfully detected and classified cube and cuboid objects within the workspace. Bounding boxes were accurately generated, and object classification enabled correct scenario selection. The detection performance was found to be reliable under consistent lighting conditions and fixed camera placement.

5.3.2 MANIPULATOR ACCURACY

The robotic arms were able to reach the detected object locations with reasonable accuracy following camera calibration and inverse kinematics computation. Pick-and-place operations were successfully executed in most trials. Minor positional deviations were observed due to mechanical tolerances and servo motor limitations.

5.3.3 MULTI ROBOT COORDINATION

The ESP-NOW-based communication framework enabled low-latency coordination between robots. The system successfully executed both handover-based and cooperative manipulation strategies without requiring continuous centralized supervision. Synchronization between robots was observed to be effective, particularly in cooperative pickup scenarios.

5.4 FAILURE CASES AND LIMITATIONS

During experimentation, certain limitations were identified:

- Misalignment of the camera affected coordinate mapping accuracy.
- Slight delays in wireless communication occasionally impacted synchronization.
- Although the defined workspace is 40 cm × 40 cm, the robot cannot pick objects placed in certain outer regions due to mechanical and kinematic constraints.Firstly, the electromagnetic (EM) gripper must remain flat and parallel to the XY plane for proper magnetic contact. This orientation requirement restricts wrist movement and prevents picking objects that require tilted end-effector positioning.Secondly, the reach of the robot is limited by the fixed length of its arm links. The maximum reachable distance is

determined by the sum of the link lengths. Objects placed beyond this radial limit cannot be accessed.Thirdly, each joint is controlled by a 180° servo motor, which restricts joint rotation between 0° and 180°. Some Cartesian positions require joint angles beyond this range, making them unreachable. Due to these combined constraints—limited arm length, 180° servo rotation limits, and flat gripper orientation—the effective pickable workspace is smaller than the defined 40 × 40 cm area, particularly near extreme edges and corners.

These limitations highlight areas for further improvement.

5.5 DISCUSSIONS

The experimental evaluation demonstrates that the proposed system is capable of performing vision-based collaborative manipulation tasks in a controlled environment. The integration of object detection, calibration, inverse kinematics, and wireless communication enabled autonomous task execution with minimal human intervention. Although the current implementation uses rule-based coordination, the system architecture is well-suited for future integration of learning-based approaches such as Multi-Agent Reinforcement Learning.

5.6 SUMMARY

This chapter presented the experimental setup and qualitative results of the proposed system. The experiments validated the feasibility of collaborative robotic manipulation using vision- based decision-making and wireless coordination.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

This chapter presents the overall summary and key outcomes of the proposed AI-powered vision-based collaborative robotic system. It highlights how the integration of computer vision, robotic manipulation, and wireless communication contributes to intelligent and autonomous pick-and-place operations in a structured workspace. The chapter reflects on the system's performance in detecting objects, coordinating multiple robotic manipulators, and executing tasks efficiently with minimal human intervention.

In addition to summarizing the contributions and achievements of the project, this chapter also discusses potential areas for future enhancement. The future scope focuses on improving system intelligence, scalability, and adaptability through advanced techniques such as learning-based coordination, enhanced perception models, and real-time optimization. These directions aim to bridge the gap between prototype implementation and real-world industrial deployment.

5.1 CONCLUSION

This project presented the design and implementation of an AI-powered vision-based collaborative robotic system for intelligent placement of goods in a warehouse-like environment. The proposed system successfully integrates computer vision, robotic manipulation, wireless communication, and coordinated task execution to address the challenges associated with handling objects of varying geometries. A custom-trained YOLOv8 object detection model was developed to detect and classify objects as cubes or cuboids using a single overhead camera. Based on the detected object type, the system dynamically selected an appropriate manipulation strategy, enabling either single- robot handover-based placement or dual-robot cooperative manipulation. The integration of camera calibration, inverse kinematics, and ESP-NOW-based communication ensured accurate and synchronized execution of robotic actions. The experimental evaluation demonstrated that the system is capable of autonomously detecting objects, coordinating multiple robots, and completing pick-and-place tasks with minimal human intervention. Despite relying on deterministic decision logic, the system architecture was designed to be scalable and adaptable, making it suitable for future learning-based enhancements. Overall, this work validates the feasibility of collaborative robotic manipulation using vision- based perception and low-cost hardware, providing a solid foundation for intelligent warehouse automation systems.

5.2 FUTURE SCOPE

While the current implementation demonstrates effective collaborative manipulation, several enhancements can be explored to improve system intelligence, robustness, and scalability.

1. **Multi-Agent Reinforcement Learning (MARL**): The system can be extended by implementing MARL techniques to enable robots to learn optimal coordination policies through interaction with the environment. Reward- based learning can improve task allocation, synchronization, and adaptability to dynamic scenarios.
2. **Visual-Language-Action (VLA) Models**: Integration of VLA models can allow high-level

task instructions to be translated into executable robotic actions. This would enable flexible task decomposition, natural language interaction, and intelligent planning.

3. **Scalability to Multiple Robots**: The framework can be expanded to include additional robotic agents, enabling large- scale collaborative manipulation in warehouse environments.
4. **Improved Grasp Planning**: Incorporating advanced grasp planning techniques and force feedback can enhance manipulation accuracy and reduce failure rates.
5. **Real-Time Performance Optimization**: Further optimization of vision processing and communication latency can improve system responsiveness for real-world deployment.
6. **Dynamic Environment Handling**: Future work can focus on enabling robots to operate in dynamic environments with moving objects and varying lighting conditions. These enhancements can significantly improve the autonomy and intelligence of collaborative robotic systems, making them suitable for real-world industrial applications.

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AI Powered Vision Task
FOR PLACEMENT OF GOODS
Two 3DOF Manipulators with Electromagnet Grippers
Centralized Camera and Main PC
ESP32 Based Peer-to-Peer and Base Communication
Simulation-Based Digital Twin (ROS2)
Scenario 1: Robot 1 Pick-and-Handovers to Robot 2
Detection & Localization
Path Planning
Pick & Drop
Drop Area
Handover
Detection
Path Planning
Scenario 2: Both Robots Pick Up the Object Together
Detection & Localization
Simultaneous Pickup
Simultaneous Pickup
Detection
Drop Area
Peer-to-Peer Communication
PC Connection
Simulation-Based Digital Twin
(ROS2)


AI Powered Vision Task
for Placement of Goods
Smart Robotic Handling System for Efficient Automation
Scenario 1: Handover Process
Detecting Object...
Cube
1 Camera Detects Object.
Handover Point
2 Robot 1 Picks & Transfers
3 Robot 1 Picks Object
Handover Point
Robot 2 Drops Object
Central Camera
Main PC & Main ESP
Base Control & Coordination
Robot 1 ESP
3DOF Manipulators with Electromagnet Grippers
Peer-to-Peer Communication
Scenario 2: Dual Pickup
Both Robots Pick & Place Together
Real-World System
Digital Twin
Synchronized Automation & Seamless Communication

AI Powered Vision Task for Placement of Goods



The AI-Powered Vision Task for Placement of Goods is an intelligent robotic automation system designed to accurately detect, pick, handover, and place objects using computer vision and coordinated robotic control. The system uses a camera-based AI vision model to identify objects, determine precise pickup locations, and guide robots through safe and reliable manipulation tasks. By integrating scene-based decision logic, the system supports multiple operational modes such as coordinated dual-robot placement and adaptive handover-based transfer.

KEY FEATURES

- AI-based object detection and localization
- Scene-based task execution (Scene 1 & Scene 2)
- Coordinated dual-robot handover
- Safe and explainable robotic operation

SOFTWARE

- Python
- OpenCV
- YOLO (AI object detection)
- Scene-based control logic
- Serial / wireles communication
- Inverse kinematics algorithms

OBJECT DETECTION





SCENE 1



SCENE 2

HARDWARE

- ESP32 microcontroller
- Robotic arm with end-effector
- Ceiling camera
- Servo motors
- Motor driver modules
- Power supply unit

SYSTEM OUTPUT

- Accurate object pickup and placement
- Reliable robot-to-robot handover

TEAM MEMBERS

JAYASRI RANI S - CSE

SAMYUKTHA SRI A - ECE

SUDHARSAN M - RAE

ERICSSON

ERICSSON - CHENNAI VISIT

LIST OF FEEDBACKS AND SUGGESTIONS PROVIDED BY ERICSSON EXPERTS

The Ericsson experts provided valuable suggestions and feedback to enhance the proposed system's real-world applicability and scalability. They emphasized the need to improve logistics efficiency and sustainability impact to make the system more deployable in industrial environments. It was recommended to clearly differentiate the system from conventional industrial robots by highlighting its unique capabilities and application scope. The experts also encouraged the development of a functional prototype with strong startup potential and suggested participation in platforms such as TN Startup to support innovation and funding opportunities. Additionally, they highlighted the importance of clearly defining real-world use cases and industrial applications of the system. The overall concept and system logic were appreciated for their innovation, and the team's approach received positive recognition. The vision-based detection system was identified as a strong component, demonstrating effective performance. The use of an electromagnetic gripper was appreciated for its simplicity and practicality. The 3D-printed robot design and fabrication process were well acknowledged, with interest shown in understanding the workflow. The collaborative coordination between robots, particularly the smooth handover mechanism, was positively noted. The attempt to incorporate a digital twin was recognized as a forward-looking feature. Furthermore, the training and implementation of cube and cuboid detection models were appreciated for their accuracy and relevance. Overall, the system integration, design approach, and potential for real-world industrial application were strongly recognized and appreciated by the experts.



Fig i:Presenting our solution to Elena Fersman Vice President and Head of AI Accelerator – Ericsson



Fig ii : Demonstration of the proposed robotic vision and collaboration system to Ericsson Experts